The constant $A(p)$

Quasi-stationary amplitude, closed forms, and the Koenigs obstruction

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Contents

Chapter 1

The constant $A(p)$

1.1 Scope

For the subcritical binary Galton–Watson process, late survival has the form

$$S_n \sim A(p)(2p)^n, \qquad 0 \le p < \tfrac{1}{2},$$

and the limiting population mean conditional on survival is $1/A(p)$. The probabilistic construction, extinction theory and conditional-moment calculation were developed in Chapter M, *Mathematical introduction*. The present chapter takes the resulting constant as its object and develops its analytic structure.

The argument is developed in layers. An exact infinite product leads to a telescoping series, two-sided and parity bounds, and a near-critical theorem with a controlled remainder. These results make it possible to compare the discrete constant with the explicit continuous-time value $A_{\mathrm{c}}(p)$ before assessing the negative closed-form search. The second half then identifies $A(p)$ with a basin value of the Koenigs linearising function for the logistic map. The Becker–Bergweiler classification justifies a precise obstruction: for each fixed $r \in (0,1)$, the function $z \mapsto \psi_r(z)$ is hypertranscendental. It does not determine the arithmetic nature of any particular number $A(p_0)$, or the differential algebraicity of the parameter map $p \mapsto A(p)$. The scope taxonomy and inverse-symbolic evidence are organised to keep those questions distinct.

The chapter closes by turning the analytic results into a practical evaluation scheme. The elementary repelling cases $r = 2$ and $r = 4$, both outside the subcritical window, are derived in the appendix.

1.2 Standing setup

Let $Z_0 = 1$, and let each individual independently leave two offspring with probability p and none with probability $1 - p$. With $S_n = \mathbb{P}(Z_n > 0)$, the offspring PGF $\phi(z) = pz^2 + (1-p)$ gives

$$S_{n+1} = 2pS_n - pS_n^2, \qquad S_0 = 1. \tag{1.1}$$

Throughout the chapter

$$\varepsilon = 1 - 2p, \qquad r = 2p, \qquad 0 \le p < \tfrac{1}{2}. \tag{1.2}$$

The endpoint value $A(0) = \frac{1}{2}$ is understood by continuity from $p \downarrow 0$.

The Kolmogorov constant is

$$A(p) = \lim_{n\to\infty} \frac{S_n}{(2p)^n}. \tag{1.3}$$

Since $\mathbb{E}Z_n = (2p)^n$, conditioning on the non-zero state gives

$$\mathbb{E}[Z_n \mid Z_n > 0] = \frac{(2p)^n}{S_n} \longrightarrow \frac{1}{A(p)}. \tag{1.4}$$

The full derivation of the conditional law and its variance belongs to Chapter M, *Mathematical introduction*; the recap here fixes only the interface needed below.

For later comparison, the continuous-time linear birth–death process with birth and death rates λ and μ, matched through the competing-clock probability $p = \lambda/(\lambda+\mu)$, has

$$A_{\mathrm{c}}(p) = \frac{1-2p}{1-p}. \tag{1.5}$$

Its derivation is also given in Chapter M, *Mathematical introduction*, and the resulting closed form provides the explicit benchmark for the discrete constant used throughout this chapter.

1.3 The constant $A(p)$

The conditioning argument in Chapter M, *Mathematical introduction*reduces the late-time law to the single constant

$$A(p) = \lim_{n\to\infty} \frac{S_n}{(2p)^n}, \qquad p \in [0, \tfrac{1}{2}), \tag{1.6}$$

whose reciprocal is the quasi-stationary mean. The results below are original to this thesis. They are developed here because the preceding quasi-stationary analysis reduces to $A(p)$, while later chapters repeatedly use the resulting numerical scheme.

Away from criticality, approximate values are readily obtained by iterating $S_{n+1} = 2pS_n - pS_n^2$ until the ratio $S_n/(2p)^n$ stabilises. Repeating the calculation over a range of p traces the discrete curve discussed in section 1.3.4. The method becomes impractical as $p \to \frac{1}{2}^-$, however, because the number of generations required for stabilisation grows without bound.

No formula in elementary functions is known for $A(p)$. The analysis begins with an exact infinite product (section 1.3.1) and an exact series whose two-sided bounds locate $A(p)$ relative to the continuous-time constant (1.5) (section 1.3.2). The series then yields the near-critical asymptotic $A(p) \sim 2(1-2p)$ (section 1.3.3) and resolves the apparent discrepancy between the discrete and continuous curves (section 1.3.4). Against that analytic background, the symbolic-regression searches in section 1.3.5 become diagnostic rather than merely negative, and the dynamical argument in section 1.4 explains the obstruction they encounter. The practical consequences are collected in section 1.5.7.

1.3.1 An infinite-product representation

Take the nonlinear map for the survival probability,

$$S_{n+1} = 2pS_n - pS_n^2, \qquad S_0 = 1,$$

and substitute $S_n = 2w_n$. This gives

$$w_{n+1} = 2p\, w_n(1-w_n) = r\, w_n(1-w_n), \qquad r = 2p, \quad w_0 = \tfrac{1}{2}, \tag{1.7}$$

which is the logistic map. Across the full range $0 \le r \le 4$, this family exhibits the period-doubling route to chaos shown in figure 1.1. Subcritical branching occupies only the interval $r \in [0, 1)$, where the origin is globally attracting. The fact that even this dynamically simple regime carries a closed-form obstruction is developed in section 1.4.

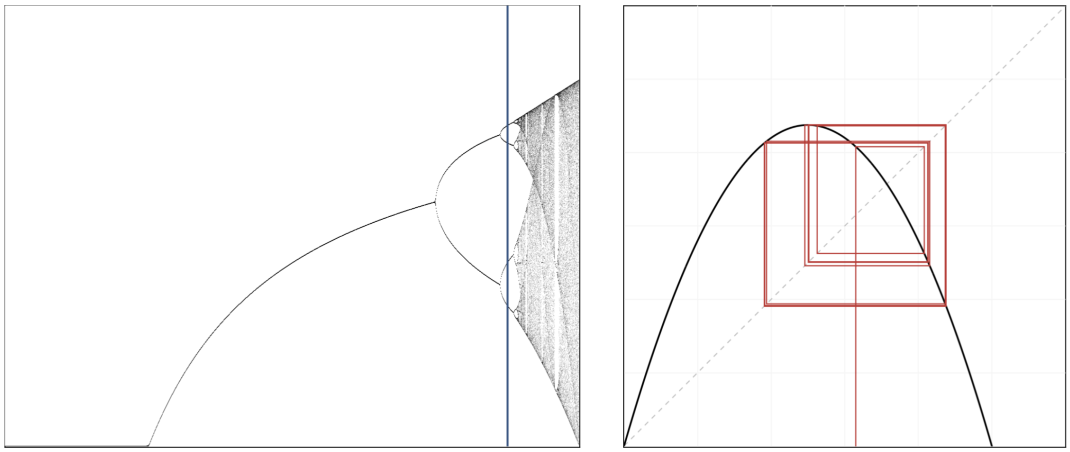

Figure 1.1: Bifurcation diagram of the logistic map $w \mapsto rw(1-w)$, showing the period doubling of stable states and the onset of chaos. The blue sample line on the left is at $r = 3.5255$; the right-hand plot shows a trajectory of the map at the same r. The regime relevant to subcritical branching is $r = 2p \in [0, 1)$, off the left-hand edge of this diagram, where the dynamics are as simple as they can be.

In terms of w, (1.6) reads

$$A(p) = \lim_{n\to\infty} \frac{S_n}{(2p)^n} = 2 \lim_{n\to\infty} \frac{w_n}{r^n}. \tag{1.8}$$

The limit in (1.8) can be reorganised as a telescoping product. For any sequence,

$$x_n = x_0 \left(\frac{x_1}{x_0}\right)\left(\frac{x_2}{x_1}\right)\cdots\left(\frac{x_n}{x_{n-1}}\right) = x_0 \prod_{k=0}^{n-1} \frac{x_{k+1}}{x_k}, \qquad \text{so} \qquad \lim_{n\to\infty} x_n = x_0 \prod_{k=0}^{\infty} \frac{x_{k+1}}{x_k},$$

Applying this identity to $x_n = w_n/r^n$ gives a ratio in which the multiplier cancels:

$$\frac{x_{k+1}}{x_k} = \frac{w_{k+1}}{w_k} \cdot \frac{1}{r} = \frac{rw_k(1-w_k)}{w_k} \cdot \frac{1}{r} = 1 - w_k.$$

Hence, with $x_0 = w_0 = \frac{1}{2}$ and $A(p) = 2 \lim_n x_n$,

$$A(p) = 2 \cdot \tfrac{1}{2} \prod_{k=0}^{\infty} (1 - w_k) = \prod_{k=0}^{\infty} (1 - w_k),$$

and pulling out the factor $1 - w_0 = \frac{1}{2}$,

$$A(p) = \frac{1}{2} \prod_{k=1}^{\infty} (1 - w_k) = \frac{1}{2} \prod_{k=1}^{\infty} \left(1 - \frac{S_k}{2}\right). \tag{1.9}$$

Convergence

For $0 < w_k < 1$, the product $\prod_{k\geq1}(1-w_k)$ converges to a strictly positive limit if and only if $\sum_{k\geq1} w_k < \infty$. In the present regime, (1.7) gives $w_{n+1} < rw_n$ for $0 < r < 1$, and induction therefore yields $w_n \leq w_0 r^n$. This geometric bound is summable, so the product converges to a positive finite value. It follows directly that the limit in (1.6) exists and that $0 < A(p) < \frac{1}{2}$ for $p \in (0, \frac{1}{2})$.

Is this an advance?

The advance is modest but useful. Equation (1.9) is exact, and its partial products decrease monotonically, so every truncation gives a rigorous upper bound on $A(p)$; the raw ratio $S_n/(2p)^n$ offers no such bound. Computing the factors still requires iteration of the nonlinear map, however, and therefore does not remove the near-critical cost. The greater value of the product is that it opens a route to the sharper series representation below.

1.3.2 An exact series and two-sided bounds

The same recursion also produces a telescoping sum, which exposes more of the dependence on p. Write $r = 2p$ and factorise the recurrence as

$$S_{n+1} = rS_n\left(1 - \frac{S_n}{2}\right).$$

Define $v_n = r^n/S_n$, so that $v_n \to 1/A(p)$ by (1.6). Then

$$v_{n+1} - v_n = \frac{r^{n+1}}{rS_n\left(1-\frac{S_n}{2}\right)} - \frac{r^n}{S_n} = \frac{r^n}{S_n}\left[\frac{1}{1-\frac{S_n}{2}} - 1\right] = \frac{r^n}{S_n}\cdot\frac{S_n/2}{1-\frac{S_n}{2}} = \frac{r^n}{2-S_n}.$$

Summing from $v_0 = 1/S_0 = 1$ gives an exact representation of the quasi-stationary mean.

Proposition 1.1 (Series representation)**.** *For $p \in [0, \frac{1}{2})$, with S_n generated by (1.1),*

$$\frac{1}{A(p)} = 1 + \sum_{n=0}^{\infty} \frac{(2p)^n}{2-S_n}. \tag{1.10}$$

Unlike the product, this representation isolates the geometric decay in a factor independent of the nonlinear dynamics; all nonlinearity is confined to the bounded denominator $2 - S_n$. Since $S_n \in (0, 1]$, that denominator lies in $[1, 2)$. Bounding it at these two extremes and summing the resulting geometric series gives the following estimates.

Proposition 1.2 (Two-sided bounds)**.** *With $\varepsilon = 1 - 2p$ as in (1.2), for $p \in (0, \frac{1}{2})$,*

$$\frac{\varepsilon}{1+\varepsilon} \;<\; A(p) \;<\; \frac{2\varepsilon}{1+2\varepsilon}, \tag{1.11}$$

that is,

$$\frac{1-2p}{2(1-p)} \;<\; A(p) \;<\; \frac{2(1-2p)}{3-4p}. \tag{1.12}$$

The lower bound is exactly half the continuous-time constant (1.5):

$$A(p) > \tfrac{1}{2}A_{\mathrm{c}}(p). \tag{1.13}$$

Proof. Since $0 < S_n \le 1$ for all n we have $1 \le 2 - S_n < 2$, and hence $\frac{1}{2}r^n < r^n/(2-S_n) \le r^n$, with equality on the right only at $n = 0$, where $S_0 = 1$. Summing over $n \ge 0$ and using $\sum_n r^n = 1/(1-r) = 1/\varepsilon$,

$$1 + \frac{1}{2\varepsilon} \; < \; \frac{1}{A(p)} \; < \; 1 + \frac{1}{\varepsilon},$$

and inverting gives (1.11). Substituting $\varepsilon = 1 - 2p$ gives (1.12), whose lower bound is $(1-2p)/(2(1-p)) = \frac{1}{2}A_{\mathrm{c}}(p)$ by (1.5). □

The two endpoints of proposition 1.2 account for the shape of the discrete curve. As $p \to 0$, one has $\varepsilon \to 1$ and the lower bound tends to $\frac{1}{2}$. The parity bound below gives $A(p) \le \frac{1}{2}$, so the lower bound is asymptotically attained and $A(0^+) = \frac{1}{2}$. At the other endpoint, $p \to \frac{1}{2}^-$, the lower bound is asymptotic to ε and the upper bound to 2ε; section 1.3.3 shows that $A(p)$ approaches the upper scale. The constant therefore moves from one bounding regime to the other across the interval, which rules out any constant rescaling of A_{c}.

Proposition 1.3 (Parity bound)**.** *For the binary Galton–Watson process started from a single individual, $A(p) \le \frac{1}{2}$ for all $p \in [0, \frac{1}{2})$, with equality approached as $p \to 0$.*

Proof. Every individual leaves either 0 or 2 offspring, so Z_n is even for every $n \ge 1$. Conditional on survival, therefore, $Z_n \ge 2$, and hence $\mathbb{E}(Z_n \mid Z_n > 0) \ge 2$ for all $n \ge 1$. Letting $n \to \infty$ and using (1.4) gives $1/A(p) \ge 2$. As $p \to 0$ the conditional population is 2 with probability tending to one, so the bound is attained in the limit. □

Table 1.1: The constant $A(p)$ computed from (1.9) in 60-digit arithmetic, together with the bounds of proposition 1.2 and the near-critical asymptotic $2(1-2p)$ from section 1.3.3. The bounds are widest in the middle of the interval and tighten at its ends: at $p = 0.05$ the lower bound is within 3% of A, whereas at $p = 0.4999$ the upper bound is within 0.1%. The final columns give the quasi-stationary mean and the number of product factors required for full precision, which is the computational cost addressed in section 1.5.7.

p	$\frac{1}{2}A_{\mathrm{c}}(p)$	$A(p)$	$\dfrac{2(1-2p)}{3-4p}$	$2(1-2p)$	$1/A(p)$	iterations
0.05	0.473684	0.486180	0.642857	1.800000	2.0568	45
0.1	0.444444	0.469382	0.615385	1.600000	2.1305	64
0.2	0.375000	0.423895	0.545455	1.200000	2.3591	113
0.3	0.285714	0.353967	0.444444	0.800000	2.8251	201
0.4	0.166667	0.237647	0.285714	0.400000	4.2079	458
0.45	0.090909	0.145069	0.166667	0.200000	6.8933	966
0.48	0.038462	0.068011	0.074074	0.080000	14.7036	2 473
0.49	0.019608	0.036395	0.038462	0.040000	27.4764	4 965
0.499	0.001996	0.003944	0.003984	0.004000	253.519	48 992
0.4999	0.000200	0.000399	0.000400	0.000400	2504.65	478 905

1.3.3 The near-critical limit

The bounds in (1.11) place $A(p)$ on the scale of ε but leave a factor of two unresolved. The exact series determines both the leading constant and the order of the first relative correction.

Theorem 1.4 (Near-critical survival amplitude)**.** *As $p \uparrow 1/2$, with $\varepsilon = 1 - 2p \downarrow 0$,*

$$A(p) = 2\varepsilon \left[1 + O\left(\varepsilon \log \frac{1}{\varepsilon}\right)\right]. \tag{1.14}$$

In particular,

$$A(p) \sim 2(1-2p), \qquad \frac{1}{A(p)} \sim \frac{1}{2(1-2p)}. \tag{1.15}$$

Proof. Write $r = 1 - \varepsilon$. The identity

$$\frac{1}{2 - S_n} = \frac{1}{2} + \frac{S_n}{2(2 - S_n)} \tag{1.16}$$

turns (1.10) into

$$\frac{1}{A(p)} = 1 + \frac{1}{2\varepsilon} + D_\varepsilon, \qquad D_\varepsilon := \sum_{n=0}^{\infty} r^n \frac{S_n}{2(2 - S_n)}. \tag{1.17}$$

Only a logarithmic bound on the remainder is needed. On the relevant range, the map $s \mapsto ps(2-s)$ is increasing in both p and s, so the subcritical survival probability is bounded by its critical counterpart, $S_n(p) \le S_n(1/2)$. At criticality, the recurrence gives

$$\frac{1}{S_{n+1}(1/2)} - \frac{1}{S_n(1/2)} = \frac{1}{2\{1 - S_n(1/2)/2\}} \ge \frac{1}{2}, \tag{1.18}$$

and hence $S_n(1/2) \le 2/(n+2)$. Since $2 - S_n \ge 1$,

$$\begin{aligned} 0 \le D_\varepsilon &\le \frac{1}{2} \sum_{n=0}^{\infty} r^n S_n(p) \le \sum_{n=0}^{\infty} \frac{r^n}{n+2} \\ &= \frac{-\log(1-r) - r}{r^2} = O\left(\log \frac{1}{\varepsilon}\right). \end{aligned} \tag{1.19}$$

Factoring $1/(2\varepsilon)$ from (1.17) gives

$$\frac{1}{A(p)} = \frac{1}{2\varepsilon} \left[1 + 2\varepsilon\{1 + D_\varepsilon\}\right]. \tag{1.20}$$

The quantity after the leading one is $O(\varepsilon \log(1/\varepsilon))$; inversion proves (1.14). □

The proof identifies the logarithmic term that slows convergence to the leading asymptotic. Numerically, the ratios $A/(2\varepsilon)$ are approximately $0.90987, 0.98612, 0.99814, 0.99977, 0.999972$ as ε decreases from 2×10^{-2} to 2×10^{-6}. An exploratory fit gives

$$1 - \frac{A}{2\varepsilon} \approx \varepsilon \left(\log \frac{1}{\varepsilon} + 0.78\right), \tag{1.21}$$

The constant 0.78 is empirical and is not used subsequently; the result needed below is the proved order estimate in (1.14).

1.3.4 Why the discrete and continuous constants differ

The preceding bounds now settle the discrete–continuous comparison. The two constants are

$$A_c(p) = \frac{1-2p}{1-p} \quad \text{(continuous time)}, \qquad A(p) = \lim_{n\to\infty} \frac{S_n}{(2p)^n} \quad \text{(discrete time)},$$

The continuous constant begins at 1, whereas the discrete constant begins at $\frac{1}{2}$; nevertheless, halving the first does not superpose the two curves.

Equation proposition 1.2 explains the discrepancy. Rather than $A = \frac{1}{2}A_c$, one has the strict inequality $A > \frac{1}{2}A_c$, with equality approached only as $p \to 0$. Combining (1.5) and (1.14), the ratio runs monotonically across the interval:

$$\frac{A(p)}{A_c(p)} \longrightarrow \tfrac{1}{2} \quad (p \to 0), \qquad \frac{A(p)}{A_c(p)} \longrightarrow 1 \quad (p \to \tfrac{1}{2}^-), \tag{1.22}$$

taking the values 0.503, 0.528, 0.619, 0.798, 0.928 and 0.988 at $p = 0.01, 0.1, 0.3, 0.45, 0.49, 0.499$. Its variation rules out any constant factor relating the two curves.

The two endpoint limits have distinct interpretations. As $p \to 0$, the factor of two is a counting effect. In the binary discrete model an individual leaves either nothing or two offspring, so a surviving population is always even (proposition 1.3); deep in the subcritical regime, conditioning on survival means conditioning on a population of exactly two, and the quasi-stationary mean is 2. In the continuous-time model, by contrast, a survivor is a single individual that has not yet died, giving a quasi-stationary mean of 1. Their reciprocals are $\frac{1}{2}$ and 1, which accounts for the discrepancy.

As $p \to \frac{1}{2}^-$, the factor disappears because both constants obey the same critical scaling. Expanding (1.5) near $p = \frac{1}{2}$ gives $A_c(p) = (1-2p)/(1-p) \to 2(1-2p)$, precisely the leading term in (1.14). At this scale the discrete generational clock no longer affects the leading order, and both models retain the same 2ε behaviour.

The discrete curve is therefore squeezed between $\frac{1}{2}A_c$ and its critical asymptote, approaching the former at one endpoint and the latter at the other.

1.3.5 Searching for a closed form

Before the connection in section 1.4 was recognised, several methods were used to search for an elementary formula for $A(p)$. Their failure is informative because it separates visual agreement from the endpoint behaviour required by the analysis above.

The first method used the continuum approximation developed in Chapter M, *Mathematical introduction*, treating the map as a differential equation in the large-generation limit. This recovers the shape of the decay but leaves its constant of integration, and hence $A(p)$, undetermined.

The second method was symbolic regression by genetic algorithm. The procedure is straightforward: generate fifty candidate functions at random, measure how closely each reproduces the target curve, retain the best five, and build a fresh population of fifty by varying those; then measure again, and repeat the cycle of variation and selection many times over [1, 2]. The first implementation searched a space of polynomials, on the hunch that the answer might take that form and because such functions are simple to encode: a genome is a coefficient list $[a_0, \ldots, a_k]$ for $a_0 + a_1 p + \cdots + a_k p^k$ up to some maximum order, drawn uniformly at random. The search quickly showed that many substantially different formulae produce curves indistinguishable from the target by eye. Supposing that a true formula might have rational coefficients, a second version searched

$\sum_i (a_i/b_i)p^i$ with $[a_i], [b_i] \in \mathbb{Z}^{k+1}$, and a third extended the space to ratios of two such polynomials. The strongest candidates fitted the high-precision numerical data closely but did not satisfy the analytic constraints. For example,

$$\hat{A}_1(p) = \frac{\frac{1}{2}(1-2p)(1+2p)}{1+\frac{1}{2}p-\frac{5}{2}p^2-\frac{6}{5}p^3+\frac{6}{5}p^4}. \tag{1.23}$$

This candidate passes through $(0, \frac{1}{2})$ and $(\frac{1}{2}, 0)$, yet fails the diagnostic from (1.14): its derivative at $p = \frac{1}{2}$ is $-2/0.55 = -3.636$, rather than the required -4.

A later attempt used a symbolic tree search over progressively nested elementary functions. It reached the same conclusion: functions of widely different form can track the target closely, even when the search explicitly favours candidates with fewer components. The returned expressions typically nested trigonometric and iterated exponential terms around a dozen or more fitted decimal constants, with no useful interpretation. One comparatively concise example is

$$\hat{A}_3(p) = \frac{-0.616647881739505\,\mathrm{e}^{-0.552605335919693\,\mathrm{e}^{p}}}{\sin(\cos p) - |p|} + 0.921964323679771. \tag{1.24}$$

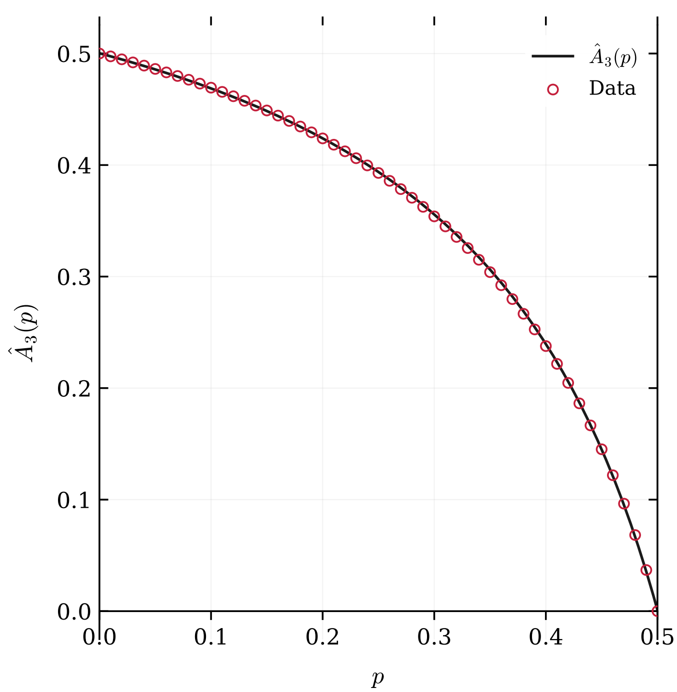

Figure 1.2: The symbolic-regression candidate $\hat{A}_3(p)$ of (1.24) (solid) against numerically computed values of $A(p)$ (red circles). The agreement is excellent to the eye across the whole interval, which is exactly the problem: the fit is not evidence of correctness. $\hat{A}_3$ misses the critical point, returning $\hat{A}_3(\frac{1}{2}) = 9.1 \times 10^{-4}$ rather than 0, and arrives there with gradient -3.63 instead of -4. Errors of this size are invisible here but are fatal for $1/A$ near criticality.

The same defect recurs across these searches. The candidates either fail to pass through $(\frac{1}{2}, 0)$ or fail to arrive there with gradient -4, and near criticality this distinction is decisive: as $A \to 0$, an error in A of vanishing absolute size produces an $O(1)$, and eventually unbounded, error in the quasi-stationary mean $1/A$. A formula may therefore appear exact on a plot of A while remaining unusable for $1/A$.

A pragmatic patch is to splice the asymptotic onto a fitted curve,

$$\hat{A}_4(p) = \begin{cases} -0.04011483983619545\, \mathrm{e}^{\mathrm{e}^{2p}} + 0.6079994336528085, & p < 0.499962, \\ 2 - 4p, & p \geq 0.499962, \end{cases} \tag{1.25}$$

which repairs the critical regime. Once a splice is admitted, however, the fitted branch is better replaced by an exact representation; section 1.5.7 develops that scheme.

These searches lead directly to the next section. The abundance of distinct formulae with nearly identical behaviour on a bounded interval is not evidence for any particular closed form. Curve fitting alone cannot distinguish a function that has such a representation from one that does not.

1.4 Relationship to the Koenigs function

The preceding recurrence already has the form of a logistic iteration. This section makes its connection with classical Koenigs linearisation precise and then uses that connection to formulate the closed-form obstruction for $r \in (0, 1)$.

1.4.1 From the survival recurrence to the logistic map

Recall that

$$S_{n+1} = 2p\, S_n - p\, S_n^2, \qquad S_0 = 1,$$

and $A(p) = \lim_{n\to\infty} S_n/(2p)^n$. Setting $S_n = 2w_n$ and $r = 2p$ yields the logistic iteration

$$w_{n+1} = f_r(w_n), \qquad f_r(w) = rw(1-w), \qquad w_0 = \tfrac{1}{2}, \tag{1.26}$$

with $r \in [0, 1)$ throughout the subcritical regime. Consequently,

$$A(p) = 2 \lim_{n\to\infty} \frac{w_n}{r^n}. \tag{1.27}$$

1.4.2 Koenigs linearisation

Definition 1.5 (Koenigs function for the logistic map)**.** Let $f_r(z) = rz(1-z)$ have multiplier r at the fixed point 0, with $0 < |r| < 1$. There exists a unique holomorphic map ψ_r, defined in a neighbourhood of 0, such that

$$\psi_r(f_r(z)) = r\, \psi_r(z), \qquad \psi_r(0) = 0, \qquad \psi_r'(0) = 1. \tag{1.28}$$

Equivalently,

$$\psi_r(z) = \lim_{n\to\infty} r^{-n} f_r^{\circ n}(z) \tag{1.29}$$

uniformly near the origin [3]. The normalisation $\psi_r'(0) = 1$ is equivalent to $\psi_r(z)/z \to 1$ as $z \to 0$.

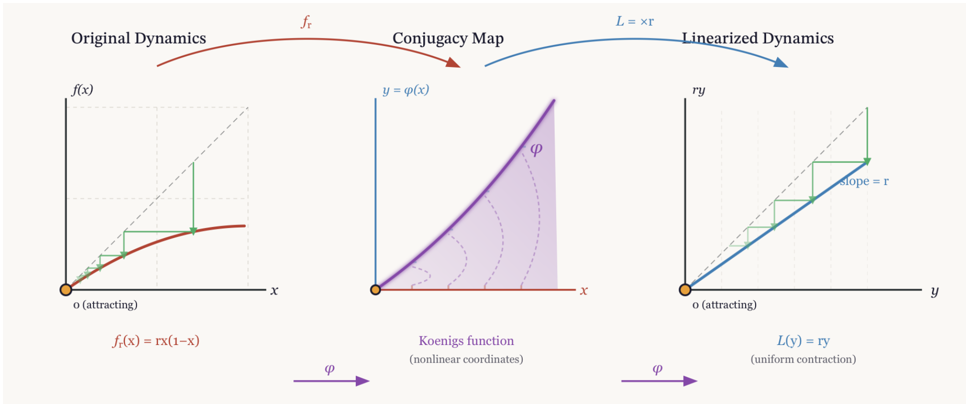

Figure 1.3: Schematic of Koenigs linearisation: f_r is conjugate, via ψ_r, to multiplication by r near the origin.

1.4.3 The identity $A(p) = 2 \lim r^{-n} f_r^{\circ n}(\frac{1}{2})$

The local Koenigs function is initially defined only near 0. For $r \in (0, 1)$, however, the origin attracts the real interval $(0, 1)$, and in particular the orbit from $w_0 = \frac{1}{2}$ satisfies $w_n \to 0$. On this basin, the limit

$$\Psi_r(w_0) := \lim_{n\to\infty} r^{-n} f_r^{\circ n}(w_0) \tag{1.30}$$

exists and agrees with the local conjugacy near 0. Indeed, $\psi_r'(0) = 1$ gives $\psi_r(w)/w \to 1$ as $w \to 0$, so $\psi_r(w_n)/r^n \sim w_n/r^n$. Once the orbit enters the local domain,

$$\psi_r(w_0) = \frac{\psi_r(w_n)}{r^n};$$

and passage to the limit yields $\Psi_r(w_0) = \lim w_n/r^n$. Taking $w_0 = \frac{1}{2}$ and using (1.27) gives

$$A(p) = 2\Psi_r(\tfrac{1}{2}) = 2 \lim_{n\to\infty} r^{-n} f_r^{\circ n}(\tfrac{1}{2}), \qquad r = 2p \in [0, 1). \tag{1.31}$$

The shorthand $A(p) = 2\psi_r(\frac{1}{2})$ will refer to this basin value of the Koenigs conjugacy, rather than to a direct evaluation of its local power series beyond the guaranteed disc of convergence.

Questions about an elementary closed form for $A(p)$ are therefore tied to the nature of the conjugacy at multiplier $r = 2p$.

1.5 No elementary closed form for the conjugacy when $r \in (0, 1)$

The obstruction rests on four short steps. First, every elementary function is differentially algebraic. Second, differential algebraicity is preserved under local compositional inversion. Third, the Becker–Bergweiler classification allows a differentially algebraic solution of Schröder's equation for a rational map of degree at least two only when the underlying fixed point is repelling. Finally, for $r \in (0, 1)$ the fixed point 0 of f_r is attracting. It follows that ψ_r is hypertranscendental at every subcritical parameter, with no exceptional case inside the attracting window. The theorem has a strict boundary, however: it does not by itself determine the arithmetic status of an individual number $A(p_0)$ or the differential algebraicity of the map $p \mapsto A(p)$. The second half of the section separates these questions before presenting the available numerical evidence.

1.5.1 Differential algebraicity, hypertranscendence, elementarity

Definition 1.6 (differentially algebraic; hypertranscendental)**.** A function g, holomorphic on a domain (or a formal power series), is *differentially algebraic over* $\mathbb{C}(z)$ if there exists a non-zero polynomial P in $n+2$ variables, for some $n \geq 0$, with

$$P(z,\, g(z),\, g'(z),\, \dots,\, g^{(n)}(z)) \equiv 0.$$

Otherwise g is *differentially transcendental*, or *hypertranscendental*.

Two closure properties make this the appropriate notion here.

Lemma 1.7 (elementary ⇒ differentially algebraic)**.** *Every elementary function in the Liouville sense — any function reachable from* $\mathbb{C}(z)$ *by a finite tower of algebraic operations, exponentials and logarithms — is differentially algebraic. Consequently a hypertranscendental function is not elementary.*

Proof sketch. Each storey of a Liouvillian tower preserves differential algebraicity. Algebraic extensions do so by elimination; if g is differentially algebraic, then the first-order defining relations for e^g and $\log g$ combine with the equation for g; and the class is closed under field operations and composition. These closure properties are classical; see Rubel's survey [4]. □

Lemma 1.8 (closure under compositional inversion)**.** *Let g be holomorphic near* 0 *with* $g(0) = 0$ *and* $g'(0) \neq 0$*, and let* g^{-1} *denote its local compositional inverse. Then g is differentially algebraic over* $\mathbb{C}(z)$ *if and only if* g^{-1} *is (Moore [5]; see also Boshernitzan–Rubel [6]).*

The inversion lemma is needed because the literature uses two opposite but equivalent conventions for Schröder's equation. Here ψ_r is the *linearising coordinate*, satisfying $\psi_r(f_r(z)) = r\,\psi_r(z)$. Becker and Bergweiler, following Ritt, classify the inverse *linearising parametrisation* $\varphi_r = \psi_r^{-1}$, which satisfies

$$\varphi_r(r\,t) = f_r(\varphi_r(t)), \qquad \varphi_r(0) = 0, \quad \varphi_r'(0) = 1. \tag{1.32}$$

By Lemma 1.8, the two conventions have the same differential-algebraic status.

1.5.2 The Becker–Bergweiler theorem, stated precisely

The required result is the 1995 classification of differentially algebraic Schröder functions [7], quoted here in the form given in Fernandes' survey of the Schröder, Böttcher and Abel equations [8, Thm 3.1].

Theorem 1.9 (Becker–Bergweiler [7]; as restated in [8, Thm 3.1])**.** *Let R be a rational map of degree at least two with* $R(0) = 0$ *and multiplier* $s = R'(0)$*, where* $s \neq 0$ *and s is not a root of unity (for* $|s| = 1$ *the Schröder function is understood as a formal series). Let φ, normalised by* $\varphi(0) = 0$ *and* $\varphi'(0) = 1$*, solve* $\varphi(st) = R(\varphi(t))$*. If φ is differentially algebraic over* $\mathbb{C}(t)$*, then:*

- (i) *the fixed point* 0 *is* repelling, $|s| > 1$*; and*
- (ii) *φ is a Möbius transformation of one of*

$$\exp(\alpha t^\rho), \qquad \cos(\alpha t^\rho + \beta), \qquad \wp(\alpha t^\rho + \beta),\ \wp^2,\ \wp^3,\ \wp'(\alpha t^\rho + \beta),$$

where $\wp$ is a Weierstrass function, $\rho \in \mathbb{Q}$ is such that the function concerned is meromorphic on $\mathbb{C}$, $\alpha \neq 0$, and β is a fraction of a period. Correspondingly R is Möbius-conjugate to $z^{\pm d}$, to $\pm T_d$ (Chebyshev), or to a Lattès map.

Remark 1.10 (lineage and independent corroboration). For $|s| > 1$, the Poincaré-function case, the classification begins with Ritt's 1926 result [9]. Becker and Bergweiler completed the picture for the three classical functional equations, and the later account by Aschenbrenner and Bergweiler [10] assigns the exceptional list explicitly to the repelling case. Di Vizio and Fernandes [11] provide independent difference-Galois corroboration: their Theorems 1.2 and 4.2 require only that s be neither zero nor a root of unity, with no assumption $|s| > 1$, and show that a differentially algebraic solution of $y(R(x)) = s\,y(x)$ in the present ψ-convention must satisfy a first-order equation of Ritt type, $(y')^\rho = B(x)y^j$. The structural conclusion used below therefore does not depend on a single formulation of the theorem.

1.5.3 Hypertranscendence throughout the subcritical window

Theorem 1.11 (no exceptional parameters in $(0, 1)$)**.** *For every $r \in (0, 1)$, the Koenigs function ψ_r of $f_r(z) = rz(1 - z)$ at 0 is hypertranscendental over $\mathbb{C}(z)$: it satisfies no algebraic differential equation. In particular, ψ_r is not an elementary function.*

Proof. Fix $r \in (0, 1)$. The degree-two rational map f_r fixes 0, where its multiplier $s = f_r'(0) = r$ satisfies $0 < s < 1$; thus the hypotheses of Theorem 1.9 hold at an attracting fixed point. If ψ_r were differentially algebraic, its local inverse $\varphi_r = \psi_r^{-1}$ would be differentially algebraic by Lemma 1.8 and would solve (1.32) with the stated normalisation. Theorem 1.9(i) would then force $|r| > 1$, contradicting $r \in (0, 1)$. Hence ψ_r is hypertranscendental, and Lemma 1.7 gives non-elementarity. □

Remark 1.12 (the exceptional set inside the window is empty, not merely avoided). The affine conjugacy $c(r) = (2r - r^2)/4$ from Lemma .14 sends $r \in (0, 1)$ to $c \in (0, \frac{1}{4})$, thereby avoiding the exceptional quadratic parameters $c = 0$ and $c = -2$, reached at $r = 2$ and at $r = 4$ respectively (with $c = -2$ also reached at $r = -2$). This parameter-plane observation is correct but not sufficient as a proof, because it does not by itself establish that the exceptional list is complete within the window. Part (i) of Theorem 1.9 removes that ambiguity: differentially algebraic Schröder functions occur only over repelling fixed points, so the attracting window contains no candidates to enumerate. Consistently, the elementary cases in Appendix .1 have the repelling multipliers 2 and 4 and are Poincaré-type linearisations, not solutions available for $r \in (0, 1)$.

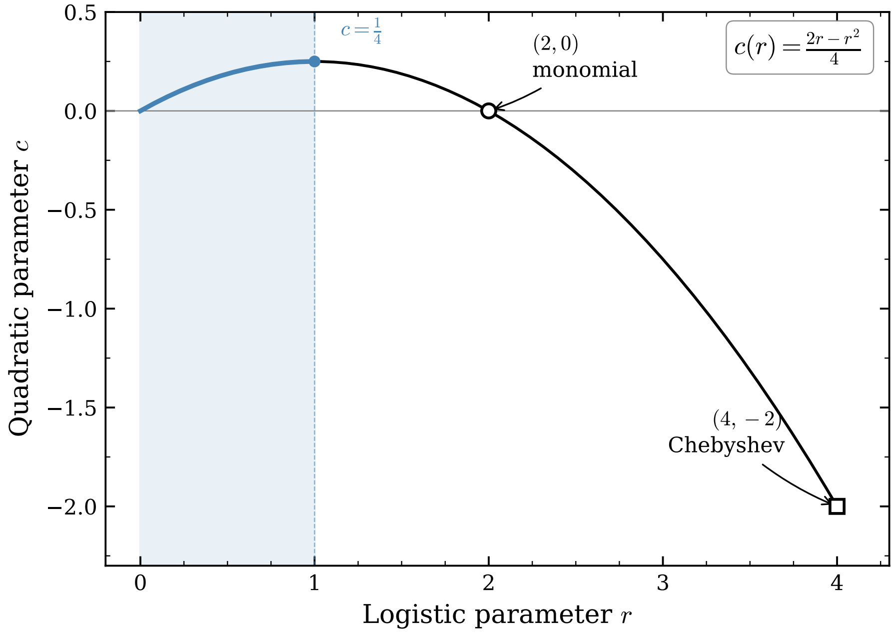

Figure 1.4: Parameter correspondence $c(r) = (2r - r^2)/4$. The segment $r \in (0, 1)$ maps to $c \in (0, \frac{1}{4})$, disjoint from the integrable values $c \in \{0, -2\}$. In the light of Remark 1.12 this picture is a consistency check rather than the proof: the exceptional linearisations live over repelling fixed points, which the attracting window excludes outright.

Remark 1.13 (germ versus basin). Theorem 1.11 concerns the analytic function ψ_r, whereas $A(p) = 2\Psi_r(\frac{1}{2})$ in (1.31) uses its extension to the real basin of attraction. For $p \geq \frac{1}{3}$, the point $\frac{1}{2}$ lies outside the disc on which the Koenigs series is guaranteed to converge, whose radius is $(1 - r)/r$. The functional equation nevertheless extends the germ by finitely many pullbacks:

$$\Psi_r(w) = r^{-N}\,\psi_r\big(f_r^{\circ N}(w)\big) \quad \text{for } N \text{ large enough that } f_r^{\circ N}(w) \text{ enters the germ,}$$

Any algebraic differential equation holding on a subdomain would persist under this analytic continuation. Hypertranscendence of the germ and of its basin extension are therefore equivalent statements, and the identity (1.31) is unaffected.

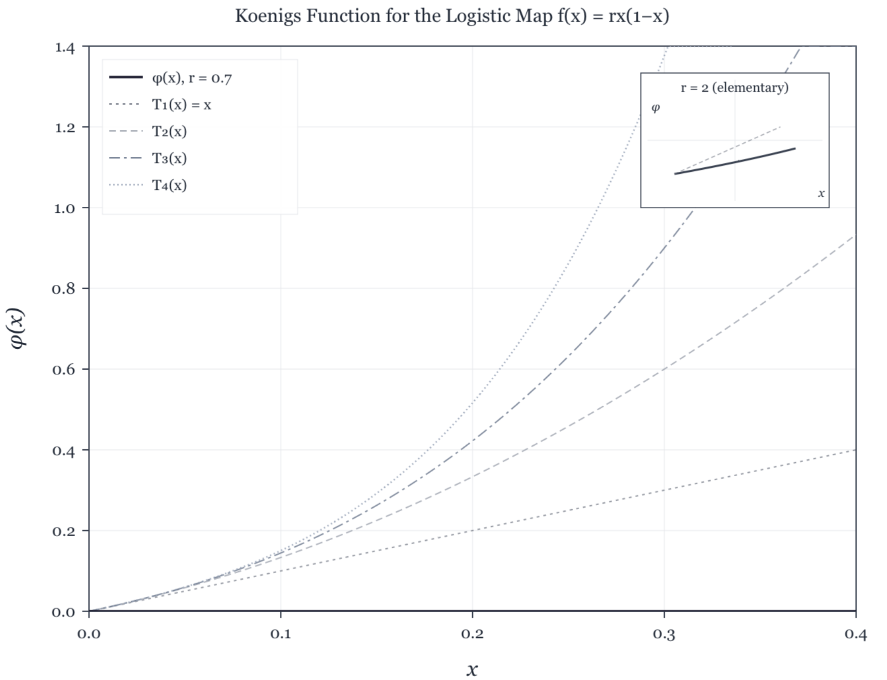

Figure 1.5: Numerical approximation of a Koenigs function for a logistic multiplier in $(0,1)$, with low-order Taylor polynomials near the origin. Elementary closed forms exist only for exceptional multipliers outside this range (Appendix .1).

1.5.4 What the theorem does and does not say about $A(p)$

The phrase "closed form" must be resolved before any claim about $A(p)$ is made. At least four inequivalent readings occur here, and a result under one reading does not transfer automatically to another:

(V) the *value* $A(p_0)$ is elementary or algebraic, for a fixed rational p_0;

(E) the *map* $p \mapsto A(p)$ is an elementary function of p;

(D) the map is *differentially algebraic in* p, i.e. satisfies some algebraic ODE in p;

(S) the map is expressible via standard special functions (Γ, ζ, ${}_2F_1$, theta or q-series).

At fixed r, Theorem 1.11 concerns the function $z \mapsto \psi_r(z)$ and establishes exactly this: *at no subcritical parameter can evaluation of $A(p)$ be reduced to an algebraic differential equation for the lineariser.* The theorem does not by itself decide any of (V), (E), (D) or (S). Two gaps remain.

The value gap. A hypertranscendental function may take elementary, or even algebraic, values at individual points. Hölder's theorem makes Γ hypertranscendental, for example, while $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. Nothing established above therefore precludes $A(\frac{1}{4})$ from being algebraic.

There is a genuine values theorem nearby: Becker and Bergweiler proved in 1993 [12] that the Böttcher conjugacy between two polynomials of the same degree $d \geq 2$ with algebraic coefficients, excluding maps linearly conjugate to monomials or Chebyshev polynomials, is transcendental and takes transcendental values at algebraic points. Their proof uses Mahler's method; see Nishioka

[13]. That theorem does not transfer to the Koenigs value $\psi_r(\frac{1}{2})$. Formally, the Schröder equation pairs an inner map of degree two with the degree-one outer map given by multiplication by r, so the equal- degree hypothesis fails. Structurally, Mahler's method balances doubly exponential analytic decay against comparable height growth. Both exponents scale like d^n in the Böttcher setting, whereas the attracting Koenigs orbit $w_n = f_r^{\circ n}(\frac{1}{2})$ decays only geometrically, $|w_n| \sim Ar^n$, while the heights of the rational w_n still grow doubly exponentially, $h(w_n) \sim C2^n$. The resulting inequalities therefore run in the wrong direction at every n. The obstruction lies in this degree structure, not in the germ/basin distinction: the pullback in Remark 1.13 moves the evaluation point arbitrarily far into the germ at the cost of a rational factor. Consequently,

transcendence — indeed, even irrationality — of $A(p_0)$ is at present not proved for a single rational $p_0 \in (0, \frac{1}{2})$.

The parameter gap. The map $p \mapsto A(p)$ lies outside Theorem 1.11, so readings (E) and (D) remain open and, to the best of our knowledge, unaddressed in the literature. The most natural machinery is the parametrised differential Galois theory of Hardouin and Singer [14], which detects differential relations in auxiliary directions for linear difference equations such as $\sigma\psi = r\psi$. It does not apply directly here because ∂_r fails to commute with the substitution operator $\sigma\colon w \mapsto f_r(w)$, which itself depends on r. The near-critical asymptotics provide no alternative obstruction: the logarithmic corrections in (1.14) are elementary functions of ε.

1.5.5 Inverse-symbolic evidence at eleven rational parameters

Because the theorems leave (V) open, the value question was examined directly by integer-relation detection. The values

$$A(p), \qquad p \in \{\tfrac{1}{10}, \tfrac{1}{8}, \tfrac{1}{6}, \tfrac{1}{5}, \tfrac{1}{4}, \tfrac{3}{10}, \tfrac{1}{3}, \tfrac{3}{8}, \tfrac{2}{5}, \tfrac{5}{12}, \tfrac{9}{20}\},$$

were computed to more than 1150 significant digits by two mechanically independent routes: iteration of the exact product (1.9), and a pullback of the orbit into the Koenigs germ followed by evaluation of the Koenigs series. The routes agreed to at least 1192 digits at every parameter. Each value was then subjected to a PSLQ battery at no less than 200-digit effective rigour:

- *algebraicity*: no integer polynomial relation of degree ≤ 10 and coefficient height $\leq 10^6$;
- *linearity*: no rational linear relation, at height $\leq 10^8$, against the constants π, π^2, $\ln 2$, $\ln 3$, $\ln 5$, γ, $\zeta(3)$, $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, e, e^π, $\Gamma(\frac{1}{3})$, $\Gamma(\frac{1}{4})$ and Catalan's constant (also tested for $1/A$, the quasi-stationary mean itself, and for A/ε);
- *bilinearity*: no representation $A = L_1/L_2$ with L_1, L_2 integer linear forms of height $\leq 10^8$ in a twelve-constant basis (run at 460 digits);
- *multiplicativity*: no relation $A = e^{q_0}2^{q_1}3^{q_2}5^{q_3}7^{q_4}\pi^{q_5}$ with rational exponents of height $\leq 10^{10}$;
- *cross-parameter*: no rational linear or multiplicative relation linking the eleven values to one another (run at 500 digits).

In all, the source record reports 123 tests with no relation and no discovery-stage candidate. Every PSLQ run terminated at its internal norm bound, so each null result means no relation was found up to the stated height. The reported precision headroom, measured as digits divided by basis size, exceeded every height cap by at least eight orders of magnitude; a returned relation would therefore have been meaningful rather than numerical noise. The present project retains this evidence as prose but does not ship the verification pipelines.

This finite evidence excludes low-complexity closed forms over the tested constants and nothing more. A form involving untested constants, such as $\Gamma(\frac{1}{7})$ or periods of higher-genus curves, or a relation above the height caps would escape the search. The calculation therefore supports the conjecture but does not prove it.

1.5.6 The working claim

For every fixed $r = 2p \in (0, 1)$, the Koenigs linearisation ψ_r is hypertranscendental in its own variable (Theorem 1.11). There are no exceptional parameters in the attracting window because differentially algebraic Schröder functions occur only over repelling fixed points. Thus no elementary expression in z for the conjugacy can be evaluated to produce $A(p)$. Whether an individual value $A(p_0)$ is transcendental, or even irrational, and whether the map $p \mapsto A(p)$ is elementary or differentially algebraic in p remain open. No closed form under any of these readings is known, and eleven rational values resist inverse-symbolic identification against standard constants at more than 200-digit rigour.

Full derivations of the elementary solutions at $r = 2$ and $r = 4$, and their place inside the Becker–Bergweiler classification, are collected in Appendix .1.

1.5.7 Computing $A(p)$ in practice

In the absence of an elementary formula, evaluation must combine exact numerical iteration with asymptotic approximation. The two regimes identified in section 1.3.2 indicate where to divide the calculation.

Across most of the interval, direct iteration is satisfactory. The infinite product (1.9) converges quickly, each partial product bounds $A(p)$ from above, and a modest number of iterations gives many digits. The difficulty is concentrated near $p = \frac{1}{2}$. As $A(p) \to 0$, the quantity of interest $1/A$ diverges, so a fixed absolute error in A becomes unbounded in the quasi-stationary mean. More importantly, the process lifetime grows sharply near criticality and S_n recedes ever more slowly. Iteration must continue until the ratio has effectively stabilised, at a cost illustrated by table 1.1: about 10^2 iterations at $p = 0.2$, 10^3 at $p = 0.45$, and nearly 5×10^5 at $p = 0.4999$.

The near-critical region can instead be handled by the asymptotic (1.14), which is exact in the limit and inexpensive to evaluate:

$$A(p) \approx \begin{cases} \dfrac{1}{2} \displaystyle\prod_{k=1}^{N} \left(1 - \frac{S_k}{2}\right), & p < p_*, \quad N \gg 1, \\ 2 - 4p, & p \geq p_*, \end{cases} \tag{1.33}$$

with the crossover p_* chosen just below $\frac{1}{2}$. Theorem 1.4 indicates where that crossover should lie. The relative error of the asymptotic is about $\varepsilon \ln(1/\varepsilon)$, so at $p_* = 0.4999$ ($\varepsilon = 2 \times 10^{-4}$) the asymptotic is already accurate to about two parts in 10^3, while the product still converges in a few hundred thousand iterations. Any p_* between 0.499 and 0.4999 gives better than one per cent accuracy throughout. If continuity of the derivative is required, the two branches may be blended across a narrow window.

This hybrid calculation produces the numerical values quoted in this chapter and in table 1.1; the corresponding conditional-mean plot appears in Chapter M, *Mathematical introduction*.

1.6 Conclusion

The constant $A(p)$ is well defined, bounded and computable, and its reciprocal is the quasi-stationary mean of subcritical binary Galton–Watson branching. The product and series representations expose complementary features of the survival recursion, while the near-critical theorem fixes its endpoint scale as $A(p) = 2\varepsilon[1 + O(\varepsilon \log(1/\varepsilon))]$.

The dynamical identification $A(p) = 2\psi_{2p}(\frac{1}{2})$, interpreted as a basin value, explains why the analytic problem is governed by Schröder linearisation. For every fixed $r \in (0, 1)$, the function $z \mapsto \psi_r(z)$ is hypertranscendental over $\mathbb{C}(z)$; the attracting window has no exceptional parameter because the differentially algebraic cases classified by Becker–Bergweiler occur only over repelling fixed points.

This theorem has a precise boundary. It proves neither irrationality nor transcendence of an individual $A(p_0)$, and it does not show that $p \mapsto A(p)$ is non-elementary or differentially transcendental in p. The eleven-parameter PSLQ study supplies only finite-height negative evidence. Thus the analytic obstruction is established for the conjugacy in its dynamical variable, while the corresponding value and parameter questions remain open.

.1 Elementary Koenigs functions at $r = 2$ and $r = 4$

These cases lie outside subcritical branching, where $r = 2p < 1$, but complete the classical picture of integrable logistic linearisations. They also locate the two elementary examples within the classification used in the main text.

.1.1 Case $r = 2$ (logarithmic)

For $r = 2$, the map is $f_2(x) = 2x(1 - x)$ and the fixed point 0 has the repelling multiplier 2. Its linearisation is therefore of Poincaré type rather than attracting-Koenigs type, but it still has an elementary form.

The affine change $u = 1 - 2x$ conjugates the dynamics to $u \mapsto u^2$. Linearising this monomial by the logarithm and returning to x gives

$$\psi_2(x) = -\frac{1}{2} \ln(1 - 2x), \tag{34}$$

normalised so that $\psi_2(0) = 0$ and $\psi_2'(0) = 1$.

Verification.

$$\psi_2(f_2(x)) = -\frac{1}{2} \ln\bigl(1 - 2 \cdot 2x(1 - x)\bigr) = -\frac{1}{2} \ln\bigl((1 - 2x)^2\bigr) = -\ln(1 - 2x) = 2\psi_2(x).$$

.1.2 Case $r = 4$ (inverse trigonometric)

For $r = 4$, the fixed point 0 of $f_4(x) = 4x(1 - x)$ has the repelling multiplier 4, so the linearisation is again of Poincaré type. Writing $x = \sin^2\theta$ gives $f_4(x) = \sin^2(2\theta)$ and therefore doubles the angle. This suggests

$$\psi_4(x) = \bigl(\arcsin\sqrt{x}\bigr)^2 \tag{35}$$

satisfies $\psi_4(f_4(x)) = 4\psi_4(x)$ on a suitable real domain, and $\psi_4(x) \sim x$ as $x \to 0$, so $\psi_4'(0) = 1$.

Verification. Using $\arcsin(2u\sqrt{1-u^2}) = 2\arcsin u$ with $u = \sqrt{x}$ (on an appropriate interval),

$$\psi_4(f_4(x)) = \left(\arcsin\sqrt{4x(1-x)}\right)^2 = \left(2\arcsin\sqrt{x}\right)^2 = 4\psi_4(x).$$

.1.3 Affine conjugacy to $z^2 + c$

Lemma .14. *The logistic map $f_r(x) = rx(1-x)$ is affinely conjugate to $P_c(z) = z^2 + c$ with $c = (2r - r^2)/4$.*

Proof sketch. Seek $h(x) = ax + b$ with $h \circ f_r \circ h^{-1} = P_c$. The substitution $x = -z/r + \frac{1}{2}$ (for $r \neq 0$) yields a monic quadratic; matching coefficients gives $c = (2r - r^2)/4$. □

Under this conjugacy, $r = 2$ corresponds to the monomial parameter $c = 0$, while $r = 4$ corresponds to the Chebyshev parameter $c = -2$; the latter is also reached at $r = -2$. Neither value lies in the image $c \in (0, \frac{1}{4})$ of $r \in (0, 1)$. As Remark 1.12 explains, this parameter count is a consistency check rather than the proof. The decisive fact is structural: both elementary cases sit over repelling fixed points, which the attracting window excludes.

.1.4 The worked examples inside the Becker–Bergweiler classification

The classification in Theorem 1.9 places every differentially algebraic Schröder function over a repelling fixed point. Such a function is a Möbius transformation of exponential, cosine or Weierstrass-$\wp$ material, while the underlying map is Möbius-conjugate to $z^{\pm d}$, $\pm T_d$, or a Lattès map. The two derivations above realise the exponential and cosine families explicitly.

Case $r = 2$. Inverting (34): $t = -\frac{1}{2}\ln(1 - 2x)$ gives

$$\varphi_2(t) = \psi_2^{-1}(t) = \frac{1 - e^{-2t}}{2},$$

an affine, and hence Möbius, image of $\exp(-2t)$: the family $\exp(\alpha t^\rho)$ with $\rho = 1$ and $\alpha = -2$. Correspondingly, $u = 1 - 2x$ conjugates f_2 to $u \mapsto u^2$, and the fixed point has the repelling multiplier 2, as Theorem 1.9(i) requires.

Case $r = 4$. Inverting (35): $t = (\arcsin\sqrt{x})^2$ gives

$$\varphi_4(t) = \psi_4^{-1}(t) = \sin^2\sqrt{t} = \frac{1 - \cos(2\sqrt{t})}{2},$$

a Möbius image of $\cos(\alpha t^\rho + \beta)$ with $\rho = \frac{1}{2}$, $\alpha = 2$ and $\beta = 0$. The fractional exponent is admissible because $\cos(2\sqrt{t})$ is entire in t. Correspondingly, $u = 1 - 2x$ conjugates f_4 to the second Chebyshev polynomial $T_2(u) = 2u^2 - 1$, and the fixed point has the repelling multiplier 4.

Both examples therefore occupy the positions prescribed by Theorem 1.9: repelling fixed points in the exponential and cosine families. For $r \in (0, 1)$ the fixed point is attracting, no classified elementary case is available, and Theorem 1.11 gives hypertranscendence of ψ_r. That distinction underpins the main-text claim for the basin value $A(p) = 2\psi_{2p}(\frac{1}{2})$.

.2 Closed-form searches and quadratic conjugacy

This appendix collects the negative and structural calculations that help interpret $A(p)$ but are not needed for its probabilistic role in the main text. The numerical formulae are approximations only. Their value is diagnostic: they show how little a visually accurate fit to $A(p)$ determines the singular quantity $1/A(p)$.

.2.1 Exploratory formulae

Polynomial and rational searches can be constrained to the known endpoints $A(0) = 1/2$ and $A(1/2) = 0$. One of the more economical rational candidates was

$$\widehat{A}_1(p) = \frac{\frac{1}{2}(1-2p)(1+2p)}{1 + \frac{1}{2}p - \frac{5}{2}p^2 - \frac{6}{5}p^3 + \frac{6}{5}p^4}. \tag{36}$$

The candidate passes through both endpoints. At the critical endpoint, however,

$$\widehat{A}_1'(1/2) = -\frac{2}{0.55} \approx -3.636, \tag{37}$$

whereas theorem 1.4 requires the left derivative -4. Matching the endpoint values while missing this slope produces an unacceptable relative error in A, and hence in $1/A$, near criticality.

A less restricted symbolic tree search produced many longer formulae with similar visual accuracy. Two representative outputs were

$$\begin{aligned}\widehat{A}_2(p) = 0.31696448624690005 \log\Bigg(\Bigg|\cos\Bigg(\bigg(\frac{1}{3.19801133824532}\bigg)^{1/4} \\ + p\Big[\sin\Big(\cos\big(|p|(0.621865696665606 - p)\cos(\mathrm{e}^{\mathrm{e}^{p}})\big)\Big) + |p|\Big]\Bigg)\Bigg|\Bigg) + 0.5982282661889831\end{aligned} \tag{38}$$

and

$$\widehat{A}_3(p) = \frac{-0.616647881739505\,\mathrm{e}^{-0.552605335919693\,\mathrm{e}^{p}}}{\sin(\cos p) - |p|} + 0.921964323679771. \tag{39}$$

These expressions have no inferential status. They record a curve-fitting exercise and are not proposed identities.

One can force the correct critical branch by splicing:

$$\widehat{A}_4(p) = \begin{cases} -0.040114839836195 45\,\mathrm{e}^{\mathrm{e}^{2p}} + 0.6079994336528085, & p < 0.499962, \\ 2 - 4p, & p \geq 0.499962. \end{cases} \tag{40}$$

This construction is a practical surrogate rather than a closed form for A. If a splice is needed, the convergent product in (1.9) is preferable on the first branch and the proved asymptotic (1.14) on the second.

.2.2 Conjugacy with the quadratic family

The logistic family $f_r(x) = rx(1-x)$ is affinely conjugate to the quadratic family

$$P_c(z) = z^2 + c. \tag{41}$$

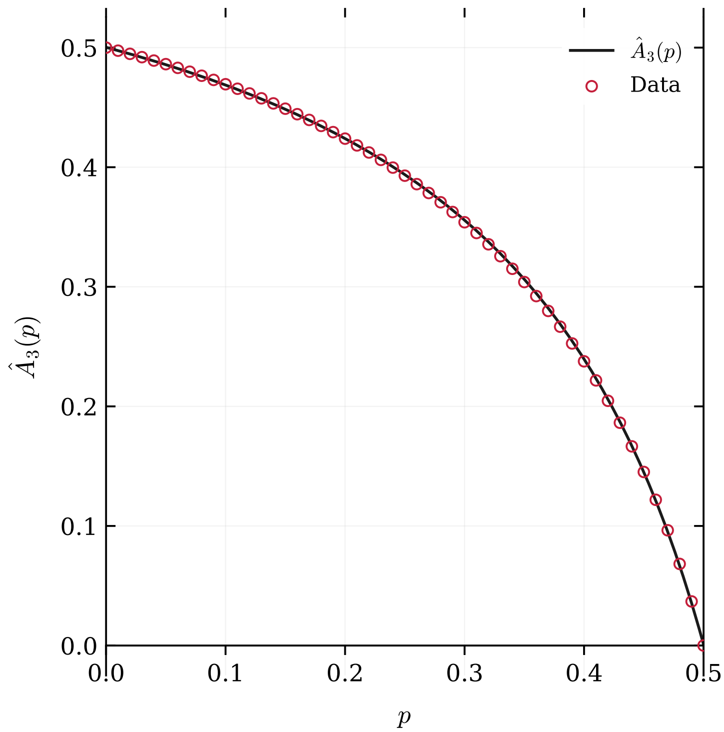

Figure 6: The fitted curve $\widehat{A}_3$ and product evaluations of $A(p)$. The discrepancy is barely visible on this scale, although $\widehat{A}_3(1/2) \approx 9.1 \times 10^{-4}$ rather than zero and its endpoint slope is approximately -3.63 rather than -4.

Set

$$z = r\left(\frac{1}{2} - x\right). \tag{42}$$

If $x' = f_r(x)$ and $z' = r(1/2 - x')$, direct substitution gives

$$z' = z^2 + c, \qquad c = c(r) := \frac{2r - r^2}{4}. \tag{43}$$

Thus the subcritical branching window $0 < r < 1$ maps onto $0 < c < 1/4$.

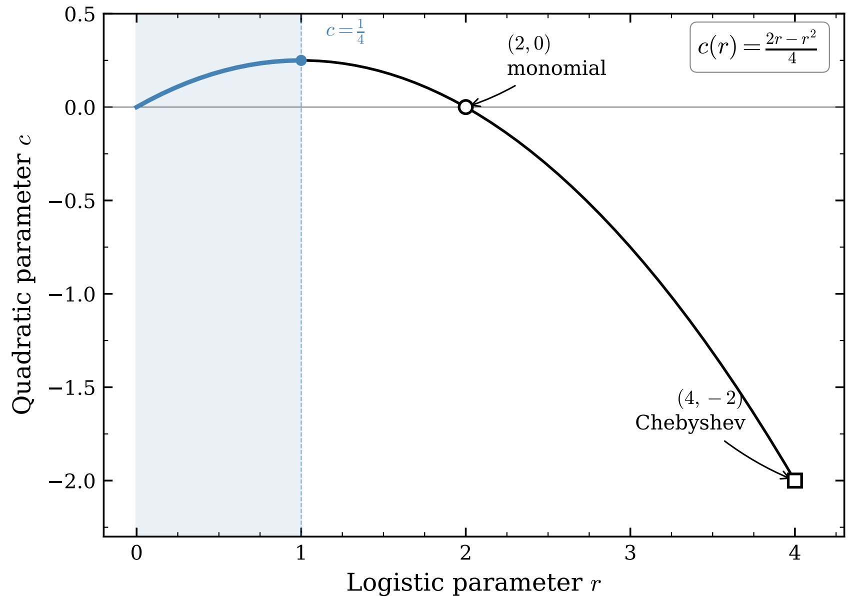

Figure 7: The parameter correspondence $c(r) = (2r - r^2)/4$. The subcritical window $0 < r < 1$ maps to $0 < c < 1/4$; the elementary monomial and Chebyshev parameters described below lie outside that window.

The interval $0 < c < 1/4$ is the real radius of the main cardioid of the Mandelbrot set. Indeed, a fixed point of $z^2 + c$ with multiplier r has coordinate $z_* = r/2$, and its fixed-point equation gives

$$c = z_* - z_*^2 = \frac{r}{2} - \frac{r^2}{4}. \tag{44}$$

Thus the endpoint $r \uparrow 1$ approaches the parabolic cusp $c = 1/4$. This observation locates the parameter range but supplies no evidence for a closed-form claim.

.2.3 The two elementary comparison cases

The exceptional quadratic maps conjugate to a monomial or Chebyshev polynomial have elementary linearisers. They provide useful algebraic checks, although neither lies in $0 < r < 1$.

For $r = 2$, put $u = 1 - 2x$. Then

$$1 - 2f_2(x) = (1 - 2x)^2 = u^2, \tag{45}$$

and the normalised local lineariser is

$$\psi_2(x) = -\frac{1}{2}\log(1 - 2x). \tag{46}$$

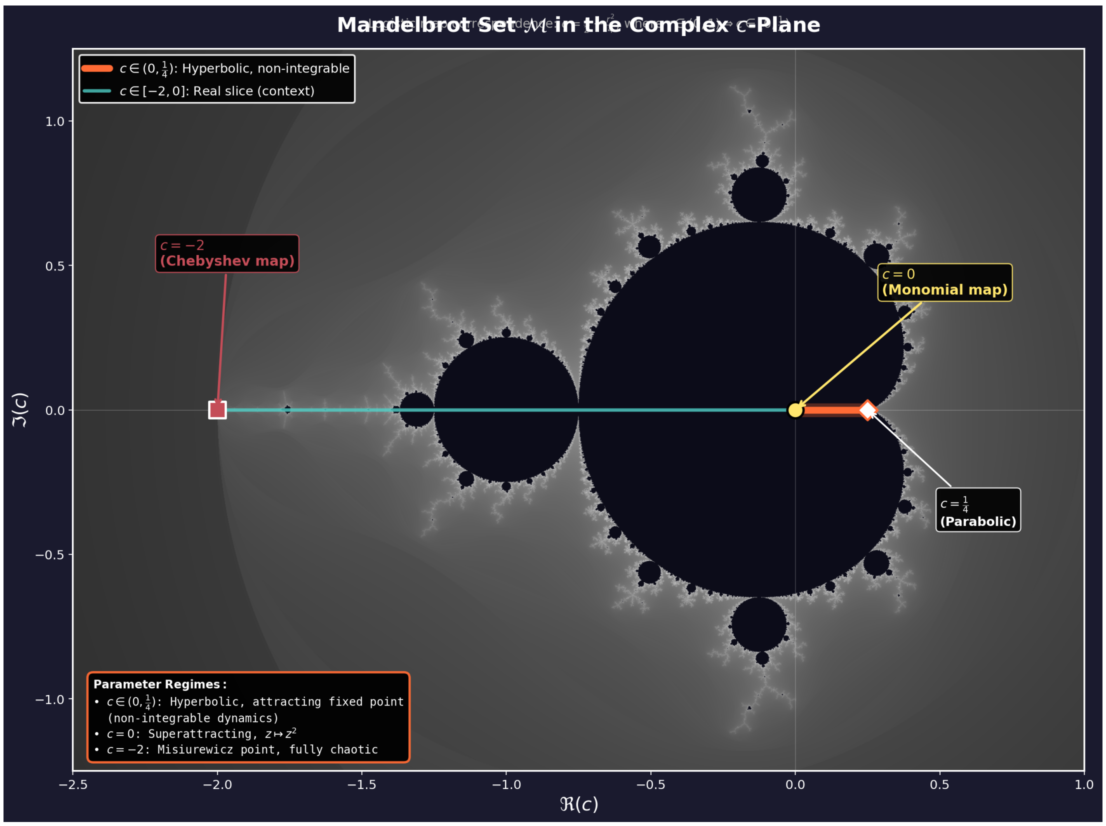

Figure 8: The image $0 < c < 1/4$ of the subcritical branching interval within the quadratic parameter plane. The exceptional parameters $c = 0$ and $c = -2$ are shown for comparison.

It satisfies $\psi_2(f_2(x)) = 2\psi_2(x)$, $\psi_2(0) = 0$ and $\psi_2'(0) = 1$.

For $r = 4$, write $x = \sin^2\theta$. The double-angle identity gives

$$f_4(x) = 4x(1-x) = \sin^2(2\theta), \tag{47}$$

so that

$$\psi_4(x) = \{\arcsin(\sqrt{x})\}^2 \tag{48}$$

satisfies $\psi_4(f_4(x)) = 4\psi_4(x)$ and $\psi_4(x) \sim x$ at the origin.

The distinction is categorical. At $r = 2$ and $r = 4$, the origin has multiplier greater than one, and the formulae (34) and (35) are local repelling linearisers. In the branching window, the origin is attracting and (1.28) defines the attracting Koenigs coordinate. The elementary formulae are therefore comparison cases, not analytic continuations of $A(p)$.

Finally, theorem 1.11 concerns $z \mapsto \psi_r(z)$ at fixed r; it does not prove that the parameter evaluation $p \mapsto 2\psi_{2p}(1/2)$ is non-elementary. Maintaining this separation is the substantive correction to the stronger claim suggested by the original closed-form search.

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